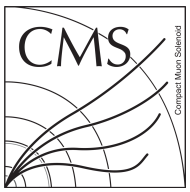
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The Compact Muon Solenoid Experiment
CMS Performance Note



Dual-threshold time-walk calibration for CMS BTL prototypes

CMS

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Abstract

A charge-dependent time-walk model is studied for CMS barrel timing layer (BTL) prototype channels using two discriminator thresholds. Their current separation divided by the measured crossing-time separation provides a finite-interval pulse-slope proxy. A linear profile fit describes this proxy as a function of the measured charge, and an inverse-slope function models the timing correction with one amplitude parameter per channel.

Slope measurements from run 4076 are compared with independent microchannel-plate (MCP) reference calibrations from run 4399. The two runs contain different events. A free constant offset and the correction amplitude are fitted only where both measurements have populated charge support. The resulting agreement is a calibration-shape comparison, not an independent measurement of timing resolution.

The draft also examines channel-to-channel variations and an exploratory prediction of the correction amplitude from internal pulse information. Such a prediction could enable operation without an event-by-event external time reference, but its transferability and full uncertainty remain to be established.

Draft v1 for scientific discussion. Not approved for public release.
The following slides retain explicitly marked review items.